Project Tutorial 1
Project Overview

Dr. Yuzuko Nakamura
Dept. of Computer Science
Y.Nakamura@ucl.ac.uk

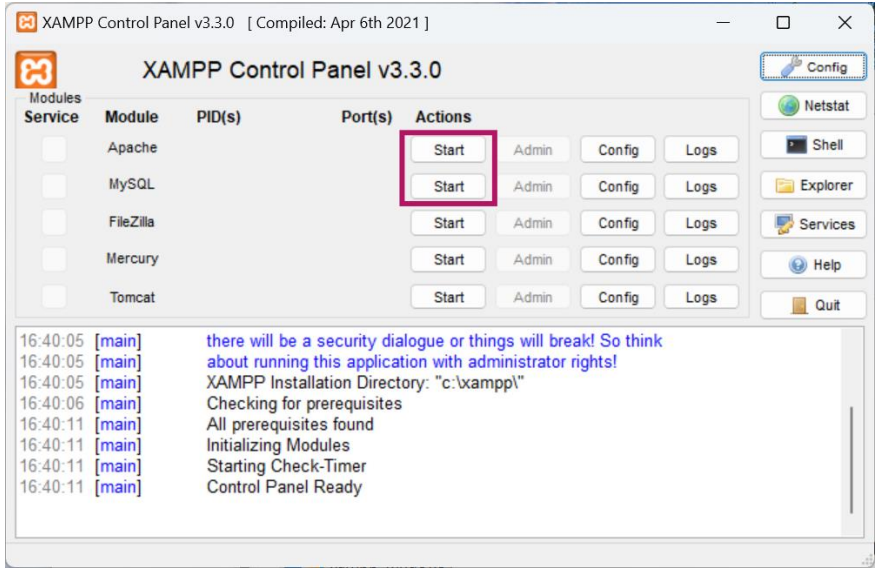
Coursework

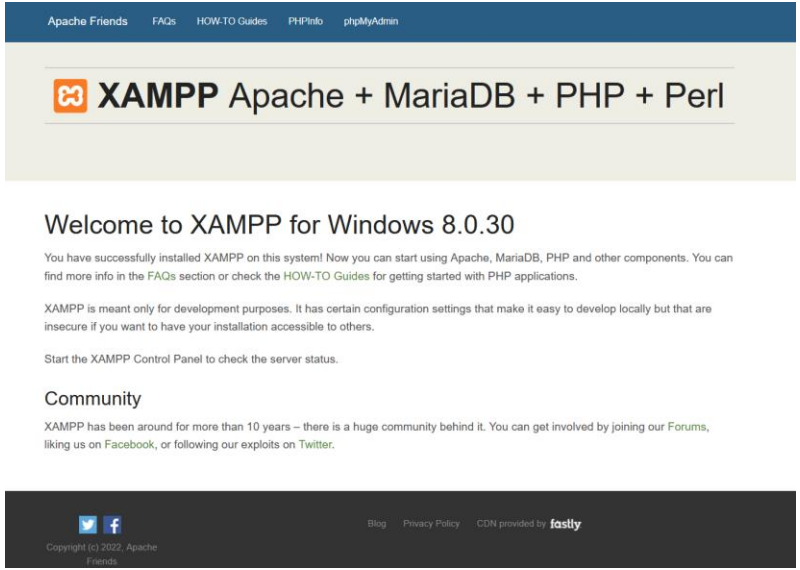
- The practical component counts for 50% of the module assessment.
- Check Moodle for coursework specification.
- Aim is to get practice designing, creating, and using a database to support a website.

Technology

- XAMPP (available on lab machines).
  - On your own computer, you can use MAMP, WAMPserver, etc.
  - **A**pache server + **M**ySQL/MariaDB + **P**HP
- Version control (e.g. Github) for group work
  - Can sign up for a Github student pack
- May want an IDE (e.g. Microsoft Visual Studio Code, IntelliJ IDEA, etc.)

XAMPP demo


XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]
XAMPP Control Panel v3.3.0
Modules
Service
Module
PID(s)
Port(s)
Actions
Apache
MySQL
FileZilla
Mercury
Tomcat
Start
Admin
Config
Logs
Netstat
Shell
Explorer
Services
Help
Quit
16:40:05 [main] there will be a security dialogue or things will break! So think
16:40:05 [main] about running this application with administrator rights!
16:40:05 [main] XAMPP Installation Directory: "c:\xampp\"
16:40:06 [main] Checking for prerequisites
16:40:11 [main] All prerequisites found
16:40:11 [main] Initializing Modules
16:40:11 [main] Starting Check-Timer
16:40:11 [main] Control Panel Ready


Apache Friends
FAQs
HOW-TO Guides
PHPInfo
phpMyAdmin
XAMPP Apache + MariaDB + PHP + Perl
Welcome to XAMPP for Windows 8.0.30
You have successfully installed XAMPP on this system! Now you can start using Apache, MariaDB, PHP and other components. You can find more info in the FAQs section or check the HOW-TO Guides for getting started with PHP applications.
XAMPP is meant only for development purposes. It has certain configuration settings that make it easy to develop locally but that are insecure if you want to have your installation accessible to others.
Start the XAMPP Control Panel to check the server status.
Community
XAMPP has been around for more than 10 years – there is a huge community behind it. You can get involved by joining our Forums, liking us on Facebook, or following our exploits on Twitter.
Blog
Privacy Policy
CDN provided by fastly
Copyright (c) 2022, Apache Friends

XAMPP

- Installs free MySQL/MariaDB software.
- Database can be accessed via web interface, phpMyAdmin.
  - Use your MySQL database as your personal sandbox!
- PHP files run differently when opened locally vs. requested from a server running PHP.

Groups

- Groups of four.
- Meet with group to decide on peer marking criteria.
- Register your group and peer marking criteria on Moodle.
- Evaluate your groupmates at end of project.
- Group project mark will be modulated by peer marks.

Deliverables

- Due date will be on Moodle.
- Group submission:
  - Video demonstrating the function of your website.
  - Design document describing project, giving link to video and including ER diagram.
  - Code files.
- Individual submission:
  - Self-assessment.
  - Peer assessment of all groupmates.

Video

- Examples on Moodle.
- Structure video with one section for each of the capabilities/features listed on the briefing sheet (in order if possible).
- Make clear how you have provided each capability/feature.
- Have a caption, or clearly state, when each capability/feature description starts.

Marking

70-100%	1st	Excellent database and design, all features working well.
60-69%	2:1	Good database and design, decent implementation of core features.
50-59%	2:2	Satisfactory database and design, some features incomplete.
0-49%	Fail	Lack of substance, issues with database design; most features not working.

Starter code

- Mostly-functional frontend made from HTML, CSS, and JavaScript.
  - Uses Bootstrap (CSS library), JQuery (JS library), and FontAwesome (icon library).
- Mostly-empty backend: no database!
  - Make your own MySQL database and use PHP's SQL functions to interact with it.

Using other libraries/frameworks

- In addition to Bootstrap and JQuery, you can use any other frontend (HTML/CSS/JS) libraries you like.
  - Using Bootstrap and JQuery not necessary.
- Do not use pre-made application frameworks like Django or a PHP framework – they do too much of the database interaction for you.
- Properly attribute any code not written by you!

Focus on the database aspect

- Projects will be marked on sophistication of database usage and design.
- Don’t spend too long on making things pretty. (No extra points awarded for sleek UIs!)

Resources

Other resources

- PHP: http://www.php.net/
- XAMPP: https://www.apachefriends.org/
- MAMP: http://www.mamp.info/en/index.html
- WAMP: http://www.wampserver.com/en/
- LAMP: use distribution package manager and/or find a tutorial on the web
- HTML: http://www.w3schools.com + many other sites.
- CSS: www.w3schools.com/css + many other sites
- MySQL: http://www.mysql.com/
  - Download MySQL Community Edition
- Bootstrap http://getbootstrap.com

UCL Lynda.com / LinkedIn Learning

- By virtue of being a UCL student, you get access to various Lynda.com tutorials / video training courses:
  - HTML/CSS
  - PHP
  - SQL
  - JavaScript
  - Bootstrap
  - and more…

Project Discussion Forum on Moodle

- Use this forum to discuss the project.
- Ask questions.
- Post useful information.
  - e.g., Useful tutorials, tools, tips and tricks, etc.
- As mentioned on Moodle, please ask general-interest questions there for quicker response.
  - Personal issues/questions can be emailed to me.

Note on AI tools

Use of AI tools is allowed

- Usage of ChatGPT and other AI tools is allowed in this class.
- However, exercise discretion when using it.
  - Plagiarism and academic integrity.
  - Limitations of AI tools.
  - Your learning.

Academic integrity

- Always properly attribute work to where you got it, whether it's an AI tool, an internet resource, etc.
- Use comments to attribute the source of code snippets.
- Avoid heavy reliance on text generation, and use quotation marks and citation if generating large blocks of text.
- Potential plagiarism issue otherwise – do not pass off other people's work as your own.
- Information on academic misconduct: https://www.ucl.ac.uk/academic-manual/chapters/chapter-6-student-casework-framework/section-9-student-academic-misconduct-procedure

Limitations of AI tools: factual accuracy

- ChatGPT and similar tools generate plausible-sounding answers to text prompts based on text they have read.
  - Very impressive and sometimes uncanny capabilities!
- However, they can make mistakes especially if those mistakes are commonly encountered ones.
- Unlike talking to a person, AI tools rarely indicate uncertainty in their responses – come off as misleadingly confident!
- Beware of fake citations!

Limitations of AI tools: writing quality

- Code generation is pretty good and generally easy to check (it either works or does not).
  - Although be careful of hidden bugs.
- ChatGPT generates grammatically correct writing but not particularly good writing.
  - Poor organization, inclusion of irrelevant information, overly wordy writing that doesn't say much, etc.
  - Your writing is probably better than ChatGPT's.

Your learning

- Use AI tools to enhance rather than detract from your learning.
- Coming up with code/logic and testing out prewritten code are two different skills.
  - Don’t entirely replace opportunities to do one with the other, especially as a new programmer.

TODOs this week:

- Introduce yourself and start finding a group.
- Read coursework write-up.
- Install XAMPP or similar on your machine.
- Experiment with setting up a MySQL database.